

Inle Myat Bush

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Education

Cornell University

2021 - 2025

B.A. in Computer Science, B.A. in Mathematics, & B.A. in Biology

Experience

Han Lab, Cornell University

02/2022 - Present

Research Assistant

- Computational and optogenetic dissection of dendrite morphogenesis
 - Developed 10+ computer vision and data analysis pipelines for analysis of time-lapse images of dendrite development (preprocessing, registration, segmentation, skeletonization, quantification, analysis and visualization) in Java, Python.
 - Coauthored 1 paper and 1 book chapter on methods developed.
- Mechanisms of synapse pruning in *Drosophila* olfactory receptor neurons
 - Designed and led investigation of cellular mechanisms and behavioral implications of excitation induced synapse pruning.

Navlakha Lab, Cold Spring Harbor Laboratory

01/2022 - Present

Research Intern

- Modeled network dynamics of neuromuscular junction connectome development. Analysis of connectome data, simulation, mathematical analysis of dynamical systems.
- Quantified and modeled network dynamics in *Arabidopsis* root systems
 - Lead quantification of time-lapse videos of *Arabidopsis* root network branching structure dynamics.
 - Formalized plant root foraging in reinforcement learning framework.

Ruta Lab, Rockefeller University

06/2024 - 08/2024

Chemers Neustein Summer Undergraduate Research Fellow

- Neuroethological study of Territorial Aggression in *Drosophila prolongata*
 - Designed a suite of ethological experiments to dissect territorial aggression and courtship. Performed optogenetic manipulations to test hypotheses on evolution of aggression.
 - Presented poster at end of summer Rockefeller University Symposium

Su Lab, Colorado University Boulder

05/2019 - 08/2019

Research Intern

- Full time summer intern in genetics and cancer research, studying tumor culling mechanisms during development in *Drosophila melanogaster*.
- Coauthored paper published in *PLOS Genetics* (Recognized as second author).

CIS Teaching Staff, Cornell University

05/2023 - Present

Teaching Assistant, Computational Genetics and Genomics

- 10hr/week for 1 semester. In person and online tutoring. Taught 2 lectures. Administered and graded computational genetics problem sets and exams.

Teaching Assistant, Design and Analysis of Algorithms

- 15hr/week for 2 semesters. In person tutoring via office hours and recitation sections. Administered and graded algorithm design problem sets and exams.

Publications and Manuscripts

1. Dasgupta S., Meirovitch Y., Zheng X., **Bush, I.**, Lichtman J., Navlakha S. (2023). A neural algorithm for computing bipartite matchings. *Proceedings of the National Academy of Science*. [doi:10.1073/pnas.2321032121](#)

2. Xu Y., Wang B. **Bush, I.**, Saunders H.A., Wildonger J., Han C. (2023). In vivo optogenetic manipulations of endogenous proteins reveal spatiotemporal roles of microtubule and kinesin in dendrite patterning. *Science Advances*. [doi:10.1126/sciadv.adp0138](#)

3. Xu Y., **Bush, I.**, Han C. (2023). Visualization and Measurement of Dendrite Arborization in Neurons. Chapter in *Neuronal Morphogenesis - Methods and Protocols*. [doi:10.1007/978-1-0716-3969-6](#)

4. Brown, J., **Bush, I.**, Bozon, J., & Su, T. T. (2020). Cells with loss-of-heterozygosity after exposure to ionizing radiation in Drosophila are culled by p53-dependent and p53-independent mechanisms. *PLOS Genetics*, 16(10). [doi:10.1371/journal.pgen.1009056](#)

Selected Projects

- **RNAFoldml** (Cornell CS 3110 Functional Programming) OCaml package that enables users to input both RNA sequences in FASTA format and a set of constraints to predict RNA secondary structure.

• **State Space Models for CRE Functional Classification** (Cornell CS 4775 Computational Genomics) Implemented and benchmarked SOTA deep learning architectures on genomic classification tasks.

• **Score!: Database and API** (Cornell CS 1998 Backend Development) Core contributor to SQL database and Flask (Python) API backend for college sports IOS app.

Technical Skills

Programming Languages: Python, R, Java, C, C++, Ocaml, Julia, Matlab, SQL, HTML/CSS, IJM, Javascript

Libraries/Frameworks: Pytorch, NumPy, SciPy, Pandas, Matplotlib, SkLearn, Flask, Jupyter, Maven, Postman, Git, JUnit, ImageJ, Microsoft Office, Dev. Tools (VS Code, Eclipse, IntelliJ, RStudio)

Bioinformatics: Sequence Alignment (BLAST, Clustal, Opfi), Databases (IMG, NCBI, PDB), UCSC, PyMol

Experimental Techniques: *Drosophila* husbandry and genetics; immunohistochemistry; tissue dissection; confocal microscopy; PCR; DNA and protein sequence analysis.

Languages: English, French, Burmese (beginner).

Honors

Chemers Neustein Summer Undergraduate Research Fellow, Rockefeller University	2024
NSF Scholar, Cold Spring Harbor Laboratory Undergraduate Research Program	2023
Grant Recipient, Einhorn Discovery Grant	2022
Best Overall Prize, Cornell AppDev Hack Challenge	2021
Sponsor Prize, Cornell BRH Hackathon	2021
Dean’s List, Cornell College of Arts and Sciences	all semesters

Selected Coursework

Biology:	Computer Science:	Mathematics:
Neuroscience	Machine Learning	Numerical Analysis of Diff. Eq.
Cell Biology	Operating Systems	Honors Analysis
Genetics	Backend Development	Probability
Biochemistry	Analysis of Algorithms	Matrix Computations
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